THE GALOIS GROUP OF AN INFINITE GALOIS EXTENSION IS THE INVERSE LIMIT OF THE FINITE GALOIS SUBEXTENSION GROUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Each lambda is realized as L_lambda = K(S_i) for a finite set S_i (lemma-finite-finitely-generated), which is what makes the directedness argument by union of finite generating sets work.

Editorial note. The label lemma-infinite-galois-limit names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-normal-closure-inside-normal (finite Galois E containing any finite K(S) inside L), lemma-finite-finitely-generated (finite extensions are finitely generated), lemma-ses-galois (restriction maps G_lambda -> G_lambda' surjective in towers of finite Galois extensions), lemma-galois-infinite (canonical surjective continuous restriction Gal(L/K) -> Gal(L_lambda/K))
- Categories, Definition categories-definition-directed-system and Section categories-section-directed-colimits: directed posets and the description of filtered colimits of fields as unions; Topology, Lemma topology-lemma-topological-group-limits (limits of topological groups commute with the forgetful functor) and topology-lemma-bijective-map (a bijective continuous map of profinite spaces is a homeomorphism); cited as standard facts of sibling chapters

Source reference. Original definition 0031: sources/stacks-chow/tags.txt, source lines 118–118. Prerequisite pointer invoked at 3234; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 04AX: sources/stacks-chow/tags.txt, source lines 5632–5632. Prerequisite pointer invoked at 3238; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BU1: sources/stacks-fields/pinned-original-source.tex, source lines 707–707. Prerequisite pointer invoked at 3229; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0BMK: sources/stacks-fields/pinned-original-source.tex, source lines 3168–3168. Prerequisite pointer invoked at 3249; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BMG: sources/stacks-fields/pinned-original-source.tex, source lines 2351–2351. Prerequisite pointer invoked at 3224; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BMH: sources/stacks-fields/pinned-original-source.tex, source lines 3042–3042. Prerequisite pointer invoked at 3245; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08YE: sources/stacks-chow/tags.txt, source lines 11337–11337. Prerequisite pointer invoked at 3267; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B20: sources/stacks-chow/tags.txt, source lines 13913–13913. Prerequisite pointer invoked at 3261; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let L/K be a Galois extension with Galois group G. Let Λ be the set of finite Galois subextensions, i.e., $\lambda \in \Lambda$ corresponds to $L/L_\lambda/K$ with L_λ/K finite Galois with Galois group G_λ. Define a partial ordering on Λ by the rule $\lambda \geq \lambda'$ if and only if $L_\lambda \supset L_{\lambda'}$. Then*

1. *Λ is a directed partially ordered set,*
2. *L_λ is a system of K-extensions over Λ and $L = \operatorname{colim} L_\lambda$,*
3. *G_λ is an inverse system of finite groups over Λ, the transition maps are surjective, and*
$$G = \lim_{\lambda \in \Lambda} G_\lambda$$
as a profinite group, and
4. *each of the projections $G \to G_\lambda$ is continuous and surjective.*

Proof. Every subfield of L containing K is separable over K (follows immediately from the definition). Let $S \subset L$ be a finite subset. Then $K(S)/K$ is finite and there exists a tower $L/E/K(S)/K$ such that E/K is finite Galois, see Lemma 0BMG. Hence $E = L_\lambda$ for some $\lambda \in \Lambda$. This certainly implies the set Λ is not empty. Also, given $\lambda_1, \lambda_2 \in \Lambda$ we can write $L_{\lambda_i} = K(S_i)$ for finite sets $S_1, S_2 \subset L$ (Lemma 0BU1). Then there exists a $\lambda \in \Lambda$ such that $K(S_1 \cup S_2) \subset L_\lambda$. Hence $\lambda \geq \lambda_1, \lambda_2$ and Λ is directed (Categories, Definition 0031). Finally, since every element in L is contained in L_λ for some $\lambda \in \Lambda$, it follows from the description of filtered colimits in Categories, Section 04AX that $\operatorname{colim} L_\lambda = L$.

If $\lambda \geq \lambda'$ in Λ, then we obtain a canonical surjective map $G_\lambda \to G_{\lambda'}$, $\sigma \mapsto \sigma|_{L_{\lambda'}}$ by Lemma 0BMH. Thus we get an inverse system of finite groups with surjective transition maps.

Recall that $G = \operatorname{Aut}(L/K)$. By Lemma 0BMK the restriction $\sigma|_{L_\lambda}$ of a $\sigma \in G$ to L_λ is an element of G_λ. Moreover, this procedure gives a continuous surjection $G \to G_\lambda$. Since the transition mappings in the inverse system of G_λ are given by restriction also, it is clear that we obtain a canonical continuous map
$$G \longrightarrow \lim_{\lambda \in \Lambda} G_\lambda$$

Continuity by definition of limits in the category of topological groups; recall that these limits commute with the forgetful functor to the categories of sets and topological spaces by Topology, Lemma 0B20. On the other hand, since $L = \operatorname{colim} L_\lambda$ it is clear that any element of the inverse limit (viewed as a set) defines an automorphism of L. Thus the map is bijective. Since the topology on both sides is profinite, and since a bijective continuous map of profinite spaces is a homeomorphism (Topology, Lemma 08YE), the proof is complete. □